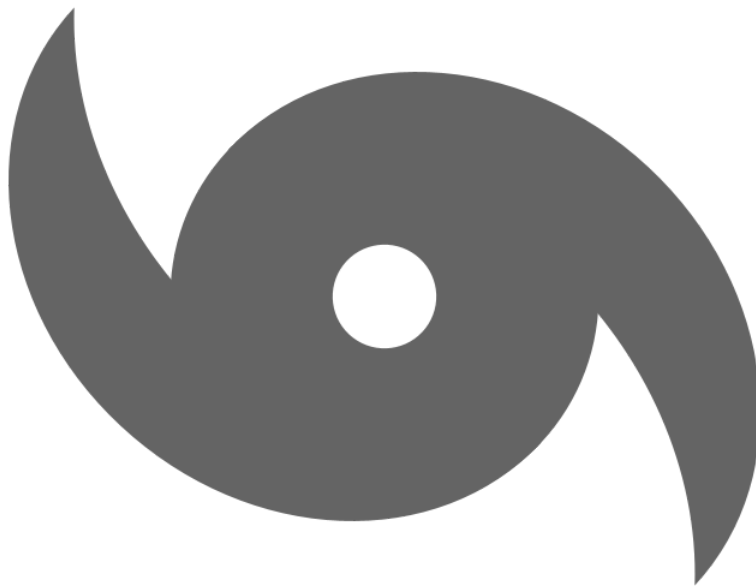


Hurricane Relief Application

Software Requirements Document



CSCE 247: Software Engineering

Group: 1 ½ Men

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v2.0

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1. Introduction

Purpose

Hurricanes strike every year, and every year their wake leaves tens of thousands in disarray. Those who are more fortunate are left unable to help their neighbors due to the lack of available communication and access to roads. Emergency services have to deal with these issues as well and are frequently overwhelmed with the surge of incoming calls. So many people working on the same problems leads to redundant information that wastes time and effort.

Our proposed solution seeks to align effective communication between emergency services and local humanitarian efforts. Using this app, friends and neighbors can organize their efforts, informing each other of the places that need the most attention, and communicating effective and safe transport routes. Professional relief organizations can make use of this information as well, gaining urgent and valuable insight on affected areas, saving and helping more people.

Scope

This document will discuss:

- The general identity of our intended audience
- Any constraints on our development of this project
- Any surface-level requirements we must fulfill
- A competitive analysis that will guide our work

2. Stakeholders

- Hurricane victims within the US
- Emergency Response and Relief workers
 - EMTs
 - Paramedics
 - Firefighters
 - Police
- Repair workers
 - Power companies
 - Internet Service Providers
 - Residential and business repairmen
 - Gas companies
- Local Governments

Personas

Jeremy Willis

Age	19
Education	High School Dropout
Profession	Unemployed
Location	Miami, Florida



Tech Savviness

Hurricane Info

Was still in high school during Hurricane Helene.

Issues with Hurricane Helene Response

“When the hurricane hit us, I was still in my house and the winds were crazy. But what really surprised me was how long it took for repairs; power and gas were out for weeks, and the roads took months to be cleared of trees.”

Herberta Ludwig

Age 28

Education High School and
Paramedic Training

Profession Paramedic

Location Myrtle Beach



Tech Savviness

Hurricane Info

Served as a paramedic in hurricane Helene.

Issues with Helene Response

“Many people died or suffered because there wasn’t adequate preparation for the hurricane; there was too little notice from authority, and too many people simply don’t care to prepare anyways. And, for weeks after the hurricane, it was hard for the people who needed help to reach us, or for us to even get to them.”

Dell Conagher

Age	42
Education	PhD in Mechanical Engineering
Profession	Contractor
Location	Houston, Texas



Tech Savviness

Hurricane Info

Volunteered to help with relief during hurricane Harvey.

Issues with Hurricane Harvey Response

“Interface with professional organizations was lacking, organization among volunteers was lacking. I’m a man who likes to solve practical problems, not deal with bureaucracy. The more time people like us who are disposed to help our neighbors spend worrying about stuff like that, the less people are getting the help they need.”

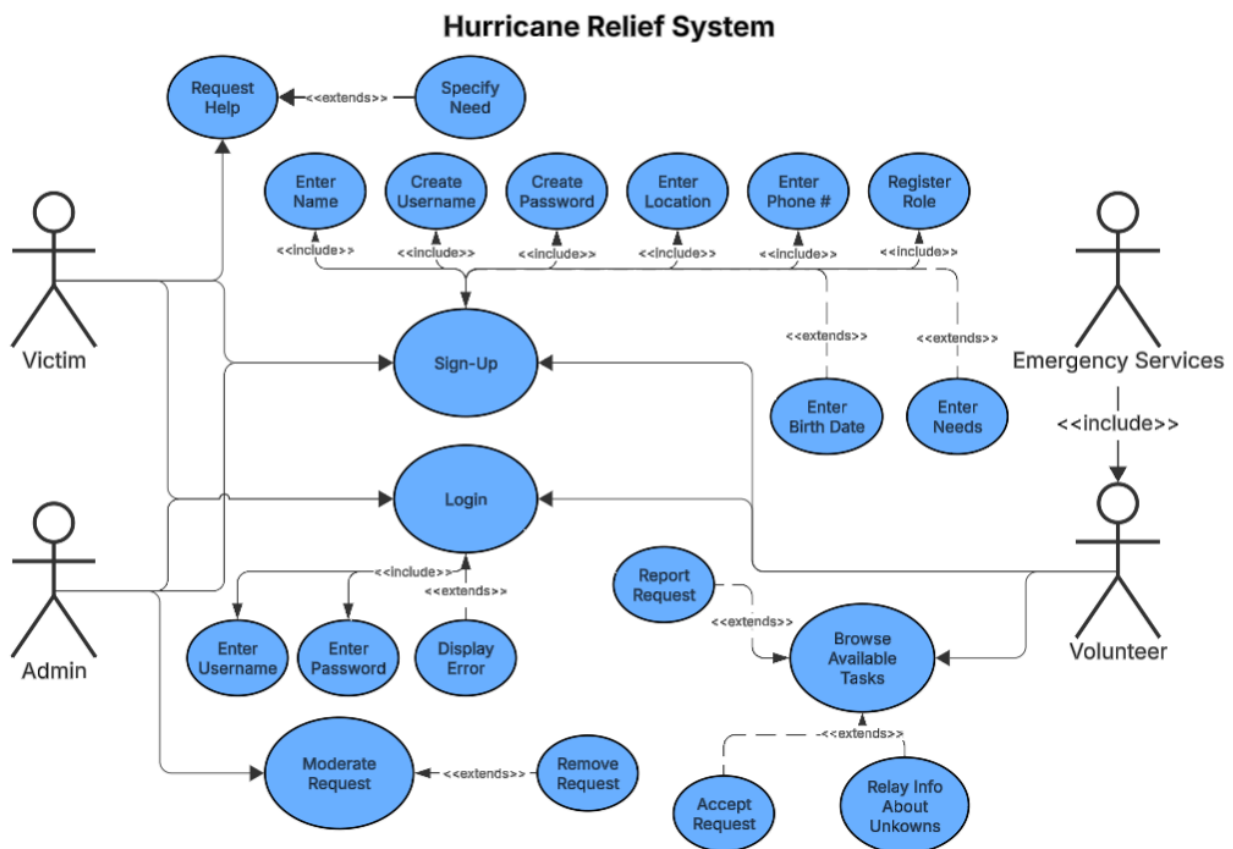
3. Constraints

- Written in Java
- Accessible on various devices
- Budget of \$0

4. Overall Description

The system must be accessible on various devices and withstand limited internet availability. A backend—if necessary—will be distributed in multiple locations to improve redundancy in case of failure due to the hurricane. When possible data should be sourced from authorities and other reputable sources and the program should prefer caution over brazenness.

5. Business Use Cases



6. Functional Requirements

https://docs.google.com/spreadsheets/d/1br4-ze_ITLnL6mW4kAi4EsiMDDtKFB1MPMoakJajW1E/edit?gid=0#gid=0

7. Non-Functional Requirements

Usability

- The app shall be usable on a smart phone
- The app shall cache info to non-volatile memory to ensure that the app remains partially functional in the case of temporary or total network loss
- The app will be intuitive for any user at any level of technological saviness
- The app will have a map of the afflicted area
- The app will clearly denote unknown areas on the map
- The app will clearly denote request locations on the map

Performance

- The app shall be power-efficient as to not waste users' battery life
- Reasonable time and space efficiency
- Should cache information to reduce network requests and save power

Security

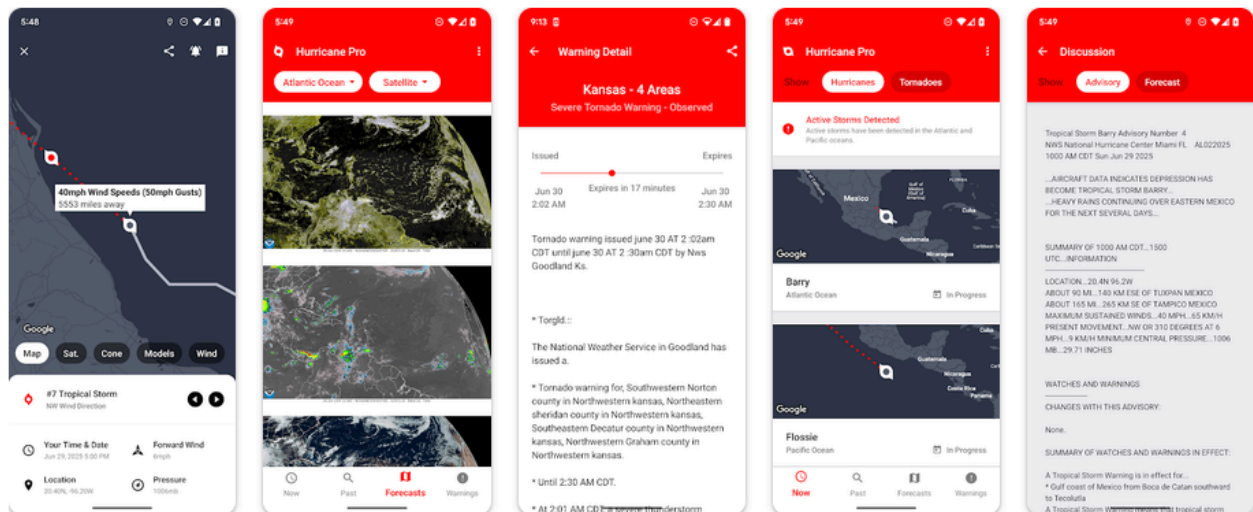
- Accounts are authenticated via username & password
- Passwords
 - Refer to the following for more information
 - [NIST Special Publication 800-63B § 5.1.1.2](#)
 - [NIST SP 800-132, Recommendation for Password-Based Key Derivation Part 1: Storage Applications](#)
 - ASCII to fit each character in one byte
 - Database should store the salt and salt + password hash
 - Use [Password-Based Key Derivation Function 2](#)
 - Salts are 128 bits (16 bytes)
 - Passwords are 8 - 48 characters
 - Iteration count of 10,000

- The resultant master key shall be 512 bits (64 bytes)

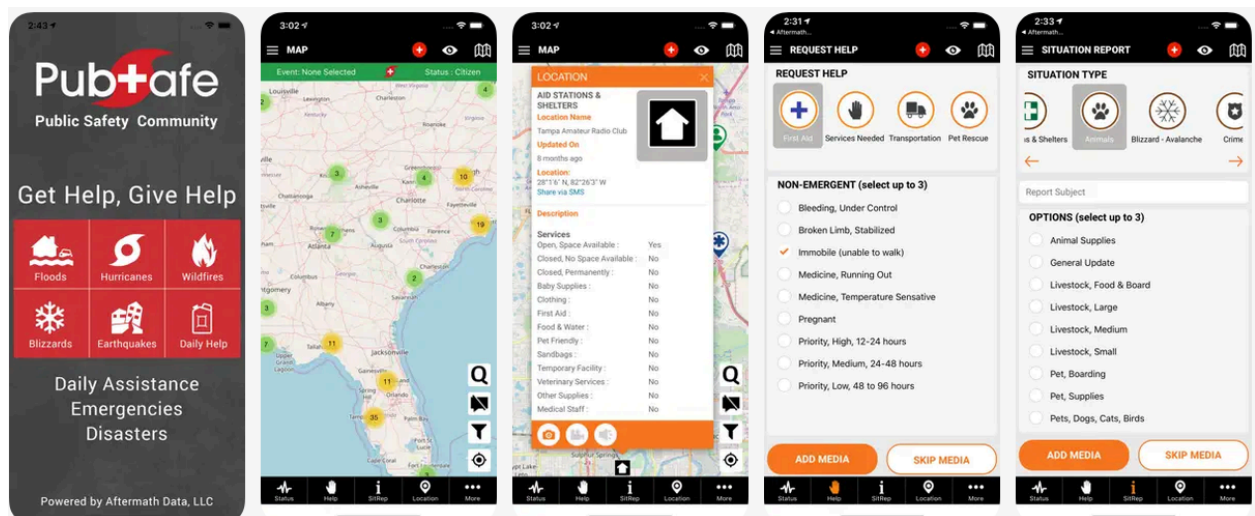
9. Competitive Analysis

App Evaluations

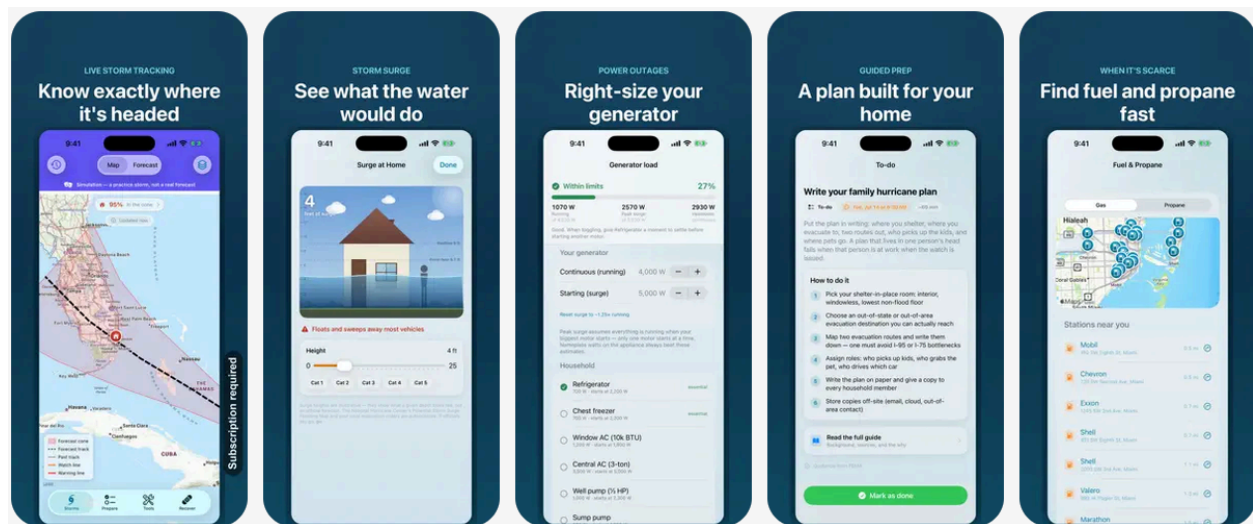
App	My Hurricane Tracker & Alerts
Strengths	● Android & iOS ● Tracks current and past hurricanes ● Displays weather service warnings ● Displays forecasts
Weaknesses	● Has a \$3.99 pro version, making certain features less accessible ● Live tracking is unavailable without internet
Audience/Focus	● People in hurricane-heavy areas ○ Primarily the Southeast US ● Focuses on preparation



App	Public Safety - Citizen & NGOs
Strengths	- Helps organize relief efforts - Works with a variety of disasters - Very self-explanatory UI - Offers a tutorial for new users
Weaknesses	- iOS exclusive - Requires registration
Audience/Focus	- Average person - Emergency services - Focuses on all disasters



App	Landfall: Hurricane Prep
Strengths	- Estimates flood levels - Estimates generator usage - FEMA-based guidance - Allows you to easily track home damage
Weaknesses	iOS exclusive
Audience/Focus	- People in hurricane-heavy areas - Primarily the Southeast US - Focuses on preparation



Summary

After analysis of various other hurricane relief-focused apps, we have discovered various successes and pitfalls that we should strive for or avoid in the creation of our project. In general, we would like to create a product with minimal limitations of who may utilize the app, avoiding any paywalls, login requirements, and reasonably low device requirements. Further, this means attempting to diminish any impact of power or internet outages on our application's functionality. These design choices are imperative for us to follow as—if we do not—our app will struggle to help with hurricane relief.

With our current plan to design an application that can help unite and organize relief efforts, we must make our application as easy-to-understand for users as possible. This includes reducing unnecessary details, including some form of guide for new users to learn the app, and many other quality-of-life additions. This goal will ensure our app will allow its users to actively engage with the hurricane relief efforts in as easy a manner as possible. Misunderstandings and ambiguity in emergency scenarios such as hurricane relief can not be afforded to those that need immediate help.